

Cryptography - 18_11

Very popular number-theoretic assumption:

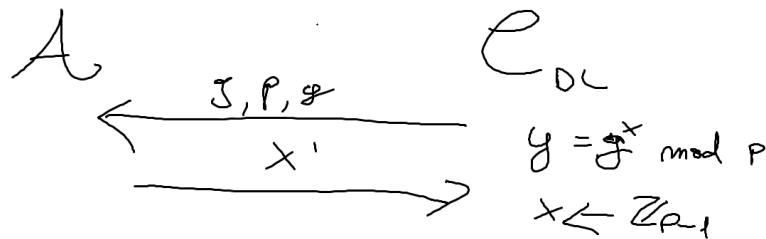
✓ Definition (DL Assumption)

Let p a prime. Then, the following holds $\forall$ PPT $\mathcal{A}$:

$$\Pr[\mathcal{A}(g, p, y) = x: x \leftarrow \mathbb{Z}_{p-1}; y = g^x \pmod{p}] \leq \text{negl}(\lambda)$$

It's the computation of the discrete log.

We can write this also as a Game:



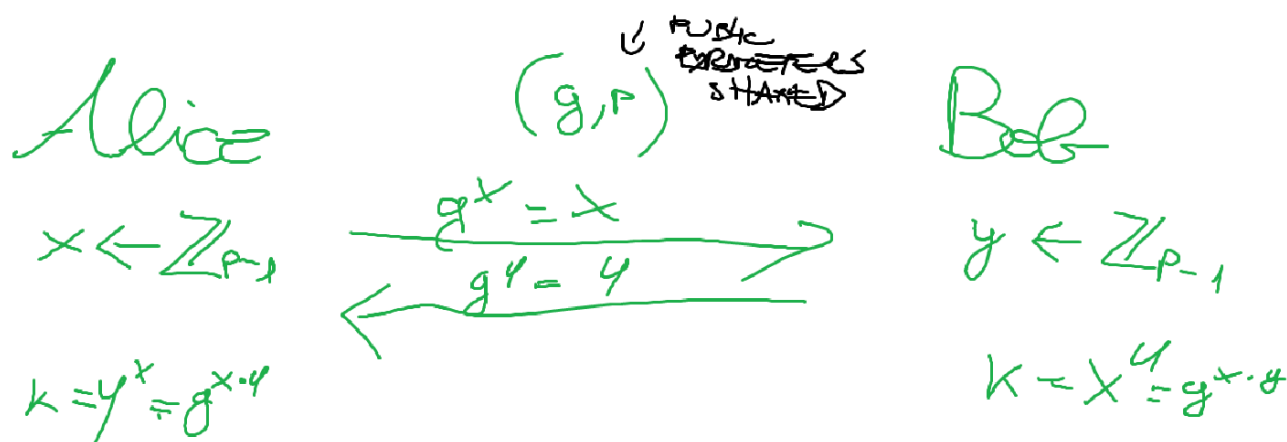
Many attempts by mathematicians (for centuries). Trivial in exponential time (we can simply try all x 's). Many better algorithms exist:

- Pollard rho,
- Baby steps Giant steps,
- Index calculus.

The complexity of all these algorithms is still sub exponential.

The public key revolution: Diffie and Hellman around 1976 proposed public key crypto.

Here is the idea:



Minimal requirement for this protocol: The DL assumption should hold.

We have two kinds of attacks:

- Passive: Eavesdrop the communication without changing it.
- Active: Can change protocol message. (Man-in-the-middle)

Security:

- Hardness of computing k
- Or better, the key is indistinguishable from uniformly random

Let's start with passive security.

We make a generalization: instead of running the protocol on $(\mathbb{Z}_p^*, \cdot)$ we now use $(\mathbb{G}, \cdot)$, where $\mathbb{G}$ is a cyclic group with generator $g \in \mathbb{G}$ and order q (i.e. $g^q = 1$ in $\mathbb{G}$).

In the exchange between Alice and Bob, we won't have (g, p) , but rather $(\mathbb{G}, g, q)$.

It is unknown if the key of the DH protocol is hard to compute for passive attackers under the DL assumption.

This motivates the following definition:

✓ Definition (CDH)

The assumption holds if $\forall$ PPT $\mathcal{A}$:

$$\Pr[\mathcal{A}(\text{params}, g^x, g^y) = g^{x \cdot y} : x, y \in \mathbb{Z}_q] \leq \text{negl}(\lambda)$$

where $\text{params} = (\mathbb{G}, g, q)$

So the probability of the adversary computing $g^{x \cdot y}$ successfully should be negligible.

Note $CDH \implies DL$. If one can solve DL it can also solve CDH .

We don't know if $DL \implies CDH$.

The CDH assumption is believed to hold in $\mathbb{G} = \mathbb{Z}_p^*$.

Now we want to have the key indistinguishable from random.

If the goal is to obtain a key that is indistinguishable from uniform, we need:

✓ Definition (DDH assumption)

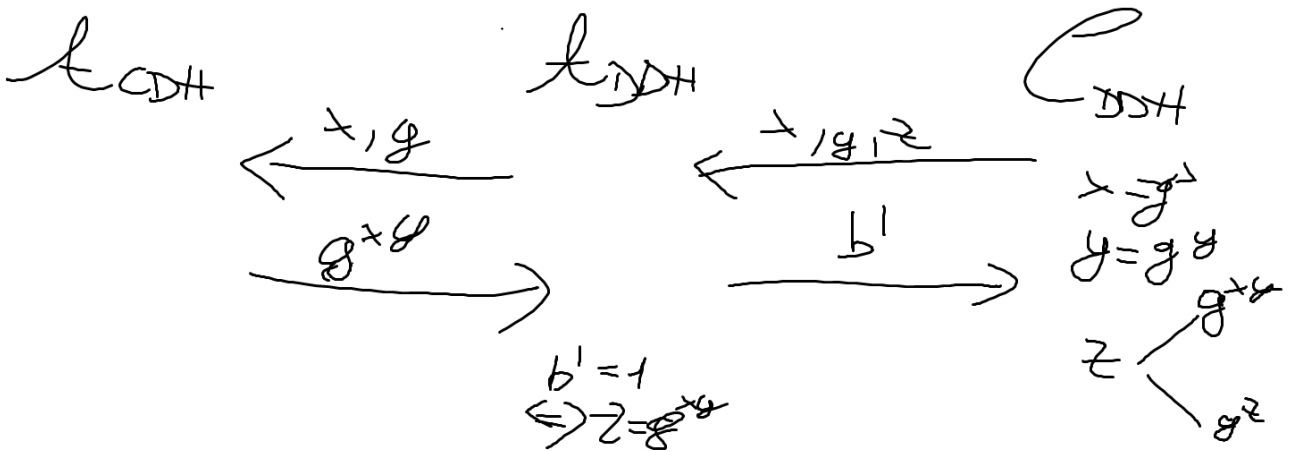
The DDH assumption holds if:

$$(\mathbb{G}, g, q, g^x, g^y, g^{x \cdot y}) \approx_c (\mathbb{G}, g, q, g^x, g^y, g^z)$$

such that $x, y, z \leftarrow \mathbb{Z}_q$.

So we can't distinguish $g^{x \cdot y}$ by g^z .

As before: $DDH \implies CDH$.



However, $CDH \not\implies DDH$ in general. Namely, $\exists \mathbb{G}$ such that CDH is believed to hold in $\mathbb{G}$, but DDH does not. For instance, $G = \mathbb{Z}_p^*$.

So, we need a different $\mathbb{G}$.

Fact

DDH does not hold in $\mathbb{Z}_p^*$.

Proof

Consider:

$$\mathbb{QR}_p = \{y : y = x^2 \text{ for } x \in \mathbb{G}\} = \{y : y = g^z \text{ for even } z\}$$

Given a modular group, consider the squares, so elements for which there exists a square root.

On the one hand, we can test if $y \in \mathbb{QR}_p$ by checking that $y^{(p-1)/2} = 1 \pmod{p}$.
Indeed, if $y = g^{2z'}$ then

$$y^{(p-1)/2} = (g^{2z'})^{(p-1)/2} = (g^{p-1})^{z'} = 1 \pmod{p}$$

.

Else, $y = g^{2z'+1}$ (the exponent is odd) and

$$y^{(p-1)/2} = (g^{2z'+1})^{(p-1)/2} = g^{(p-1)/2} \cdot \underbrace{(g^{p-1})^t}_1 = g^{(p-1)/2} \neq 1 \pmod{p}$$

This gives a distinguisher:

- If $Z = g^z$, then Z is a square with probability $\frac{1}{2}$
- If $Z = g^{x \cdot y}$, then Z is a square with probability $\frac{3}{4}$ (either g^x is a square or g^y is a square, or both)

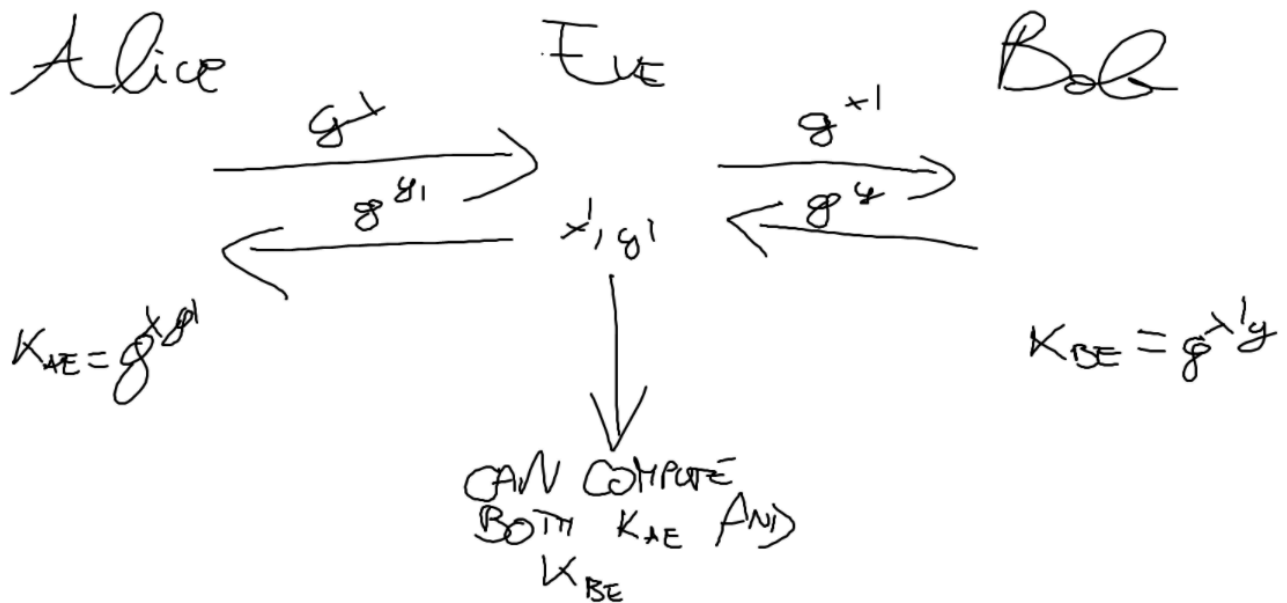
$\implies$ Can break DDH with probability $\frac{3}{4} - \frac{1}{2} = \frac{1}{4}$.

■

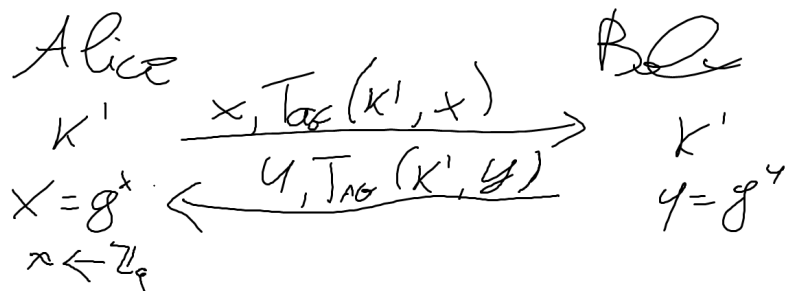
Luckily, there are many $\mathbb{G}$'s where DDH is believed to hold:

- Easiest fix: Let $\mathbb{G} = \mathbb{QR}_p$, where $p = 2q + 1$, and p, q primes. This is cyclic, with order $\frac{p-1}{2} = q$.
- Elliptic curve groups. Those are groups defined by the points of an elliptic curve, so something like $Y = X^3 + aX^2 + bX + c$, modulo a prime. So $\mathbb{G} = E(\mathbb{Z}_p)$, "+".

What about active security?

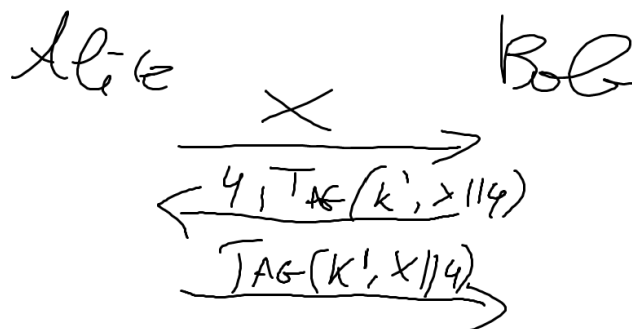


Bottom line: We need master keys to authenticate communication using either MACs or digital signatures.



This is a bad choice, an ephemeral leakage of x allows long-term impersonation.

A better protocol:



This is still not secure.

Let's show that DDH easily implies what we've studied so far:

For instance, PRGs.

Let's take $(\mathbb{G}, g, q) = \text{params}$, and $x, y \leftarrow \mathbb{Z}_q$. Now:

$$G_{\text{params}}(x, y) = G(x, y) = (g^x, g^y, g^{xy})$$

with $G : \mathbb{Z}_q^2 \rightarrow \mathbb{G}^3$, so it has positive stretch. Also $(g^x, g^y, g^{xy}) \approx_c (g^x, g^y, g^z)$, that is uniform over $\mathbb{G}^3$.

We can also improve the stretch:

$$G(x, y_1, \dots, y_t) = (g^x, g^{y_1}, g^{xy_1}, \dots, g^{y_t}, g^{xy_t})$$

with $G : \mathbb{Z}_q^{t+1} \rightarrow \mathbb{G}^{2t+1}$, and $x, y_i \in \mathbb{Z}_q$.

I'm re-using the same x .

🔗 Exercise

Prove this is a PRG under DDH.

Let's see PRFs.

Apply GGM with a particular PRG. We get:

$$\mathcal{F} : \{F_a : \{0, 1\}^n \rightarrow \mathbb{G}\}_{a \in \mathbb{Z}_q^{n+1}}$$

$$F_a(x_1, \dots, x_n) = (g^{a_0})^{\prod_{i=1}^n a_i^{x_i}}$$

The above is GGM with following PRG:

$$G_a(g^b) = (g^b, g^{ab}) = G_0(g^b) || G_1(g^b)$$

